

DISCRETE ENERGY AND COMPLEXITY

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ABSTRACT. The Hausdorff dimension of a set can be detected using the Riesz energy. Here, we consider situations where a sequence of points, $\{x_n\}$, “fills in” a set $E \subset \mathbb{R}^d$ in an appropriate sense and investigate the degree to which the discrete analog to the Riesz energy of these sets can be used to bound the Hausdorff dimension of E .

After that, we will look at applications of discrete energy to questions of learnability. In particular, we will look at connections between discrete energy and the fat-shattering dimension of a family of functions.